Technische Hochschule Nürnberg

Fakultät Informatik

RaspberryPi CyberSec Lab: Development of a Penetration Testing Platform

Bachelorarbeit im Studiengang Elektro- und Informationstechnik

vorgelegt von

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Kurzdarstellung

Diese Arbeit befasst sich mit der Entwicklung einer Testplatform für Cyberangriffe mit dem Fokus auf Funkverbindungen. Im Rahmen den Projekts wurde auf Basis von RaspberryPi OS Software entwickelt, um Umgebungen zu schaffen, in denen verschiedene Cyberangriffe getestet werden können.

Abstract

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Chapter 1.

Introduction

You may have read about similar things in [**?**]. You can also write footnotes.[1]

1.1. This is an Important Section

It is possible to reference glossary entries as library as an example.

1.1.1. And an even more important subsection

[1] Footnotes will be positioned automatically.

Chapter 2.

Basics

2.1. Cybersecurity

Through the progression of digitalization, Cybersecurity is of ever increasing relevance to the security of IT-Systems. An increasing number of devices are connected to the internet, making them part of the cyberspace and therefore potentially vulnerable to cyberattacks. Almost all sectors of our lives are at risk, when our homes, infrastructure, media devices, industries and governments are connected to the internet.

Cybersecurity aims to protect these devices by achieving the well-known CIA triad, which is an acronym for Confidentiality, Integrity, and Availability. [Oriy 17, chapter 3]

[Sank 21, page 20]

One of the earliest and most commonly used forms of attack used to target access control is a technique known as war driving. This technique, simply put, is to use a wireless enabled device along with specialized software used to detect or probe wireless networks that have come with in range of the wireless device. What makes war driving so popular and so effective is the fact that many computer users, both personal and business, have been deploying 802.11 wireless access points for years with little regard for security. The deployment of a wireless network that allows the user to roam has taken precedence over taking measures to secure their access points and devices against potential attack. In response to this deployment of wireless networks that are misconfigured or insecure, we have a category of attackers that engage in this practice known as war driving. Those who engage in this activity will build custom rigs built out of a combination of hardware and software, with the intent of cruising public venues to find both wireless access points and devices in some cases. These individuals may just be interested in locating and targeting a specific network for their own ends or, in some cases, they may even share this information to an online database or website where any visitor can search for these access points.

When working with any type of network or information one of the fundamental components that needs to be addressed is that of integrity. The integrity of data is essential as the

receiver of a piece of information needs to have confidence that what they are receiving and relying on is faithful to the intent of the creator and sender of that data. If the integrity, and therefore confidence, in data is lost then data becomes essentially worthless. If you have been in security long enough you have undoubtedly run into the well-known CIA triad, which is shorthand for Confidentiality, Integrity, and Availability.

[Oriy 17]

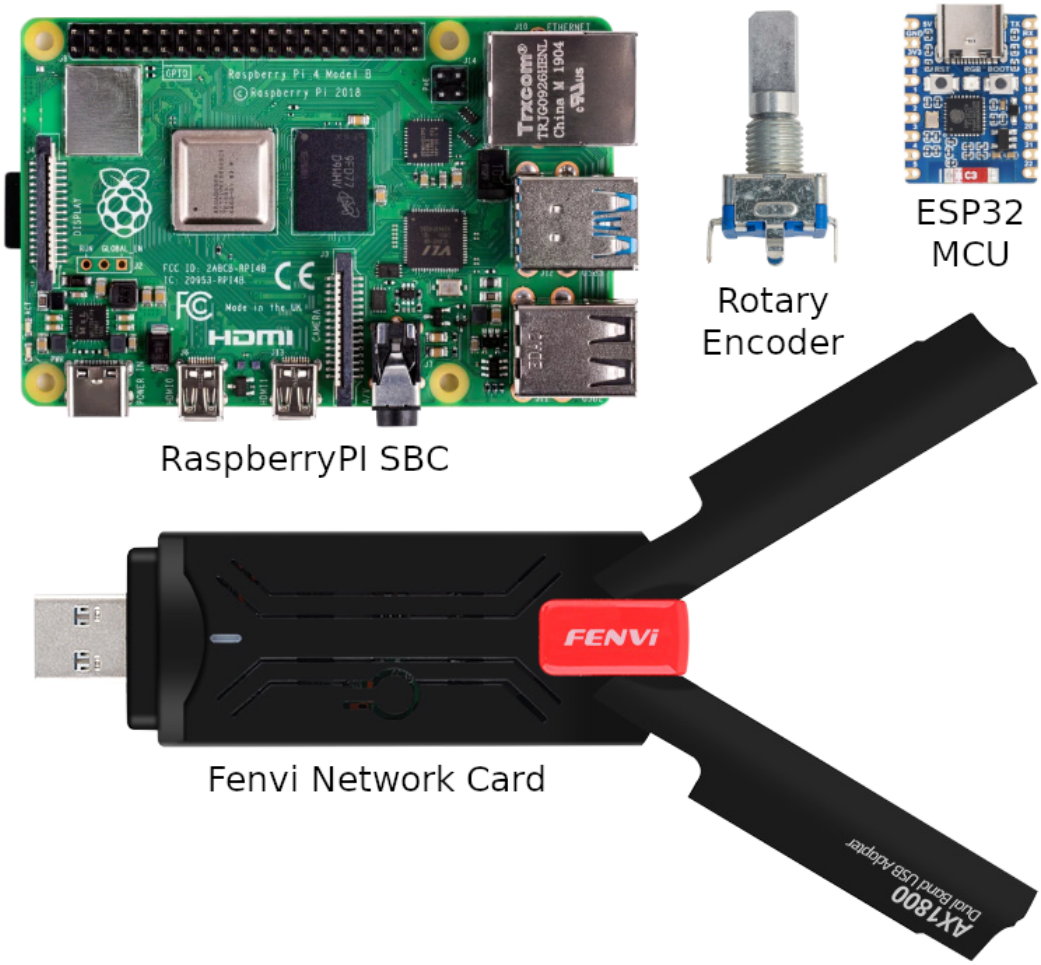
Chapter 3.

Architecture

3.1. Hardware

The Project is realized using a RaspberryPI as the central device, connected to which are an external network card, an ESP32 microcontroller, a display and a rotary encoder for user input.

As the main computer the RaspberryPI 4B with 8GB of RAM was chosen due to its ease of use, linux based operating system and widely available documentation.



3.2. Software

The software of the project is mostly modular, it consists of a central program, which runs the user interface and then executes various shell scripts, with which the functions are implemented. When executing the setup script, the main program, which is written in C++, gets compiled and the executable gets called by the start.sh script. The program then displays the user interface, which is a simple interface, printed into the console. Each page of the UI is a "Menu" object, each menu has a title, options of the type "MenuOptions", a variable to track the currently selected option, a method to add options to a menu and methods for navigation, input capture and displaying. MenuOption objects have a name, as well as an action of the std::function type, which can be executed by calling the execute() method. In the main of the program, the constructors for menues and the addOption() methods are beeing used, to create the different menues and their submenues. The program starts by call-

ing the navigate() function of the main menu, which then displays all available submenues and menu options. All submenues work in the same way as the main menu, getting their respective navigate() called upon selection. During navigation the UI peridiodically prints the available options, using unicode to highlight the currently selected option as well as to give contrast to the title of the menu.

For input the device uses a rotary encoder with a pushbutton, turning the encoder moves the selection up or down in the menues, the pushbutton executes the selected option. The encoder is connected to the RaspberryPIs GPIO interface and the input signals are read and processed by the encoder program. The start.sh script starts both programs and pipes the stdout of encoder into the stdin of main. When an action of the encoder is detected, the according command gets sent to main, the commands beeing the unicodes for the up/down arrows and the enter key on a keyboard. For example, the User turns the the encoder one click clockwise, therefore main receives one downwards arrow symbol and the selections moves to the next option below. Because no special commands are utilized for navigation, the program can be easily tested on any computer by navigating with the keyboard.

When an action is executed, main calls the according shell script with the system() function. The wifiActivate.sh script is used to open wifi hotspots, where the user can choose the type of Wifi standard beeing used. It receives an input command and then establishes a WEP, WPA, WPA2 or WPA3 network.

Listing 3.1: extract of wifiActivate.sh

```
#!/bin/bash

scriptdir="$(dirname␣"$0")"

read -n 4 connection

if [ "$connection" == "WEP" ]; then
   exists=$(nmcli connection show | sed -n '/Wifi-WEP␣/p')
   if [ -z "$exists" ]; then
      sudo nmcli device wifi hotspot con-name "Wifi-WEP" ssid "WEPtest" password
```

With the wifiReset.sh script, the user can deactivate the active network. Once a network has been cracked, the wifiNewPassword.sh script can be used to give a new random password to the selected network. The script the pulls a random password out of a password list, which contains popular passwords that have been filtered to work with the specification of each network standard. To display the current status of the network, meaning its security standard, SSID and password, the wifiStatus.sh script is used.

```
#include <string.h>
int x = 1;
```

```
if(x == 1){

    printf("x is 1.");
}
```

Listing 3.2: This is an example of inline listing

You can also include listings from a file directly:

```
x = 1
if x == 1:
    # indented four spaces
    print("x is 1.")
```

Listing 3.3: This is an example of included listing

Chapter 4.

Worklog

4.1. Abc

Chapter 5.

Wifi Attacks

5.1. Abc

Chapter 6.

Penetration Testing

6.1. Abc

Chapter 7.

Educational Use

7.1. abc

Chapter 8.

Outlook

Chapter 9.

Summary

Appendix A.

Supplemental Information

Anleitung zum Raspi-CyberSec-Lab

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6. März 2025

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[Oriy 17] S.-P. Oriyano. *Kali Linux Wireless Penetration Testing Cookbook: Identify and assess vulnerabilities present in your wireless network, Wi-Fi, and bluetooth enabled devices to improve your wireless security.* Packt Publishing, 2017.

[Sank 21] S. G. Sankaran and S. R. Gulasekaran. *Wi-Fi 6: Protocol and Network.* Artech House, 2021.

Glossary

library A suite of reusable code inside of a programming language for software development. i, 1

shell Terminal of a Linux/Unix system for entering commands. i